Andrew Makarevich

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Education

University of Maryland, College Park **Expected Dec 2028**
B.S. in Computer Engineering, Minor in Chinese · GPA: 3.69/4.00 *College Park, MD*

Experience

Systems Engineering Intern **Jun 2026 – Aug 2026**
Booz Allen Hamilton *Annapolis Junction, MD*

- Designed, fabricated, and validated a multi-signal wire harness interfacing two independent hardware systems, verifying continuity, signal integrity, and power delivery under load.
- Co-developed an automated firmware deployment tool for Industrial Control System (ICS) devices across a Purdue-model segmented network with custom subnetting and firewall rules.
- Built C/libcurl middleware for HTTP packet transmission to embedded targets, paired with a Dockerized SQLite backend and React/D3.js telemetry dashboard.

ADCS & Flight Software Engineer **Oct 2026 – Present**
TERP RAPTOR (UMD CubeSat Mission to Asteroid Apophis) *College Park, MD*

- Design the flight software (FSW) interface to a commercial ADCS for a 2029 Apophis flyby CubeSat, defining the serial command and telemetry link and the FSW modules that use it.
- Translate ADCS interface documentation into FSW command, telemetry, mode, and fault-handling requirements for the Preliminary Design Review.
- Specify FSW checkout support for reaction wheels and attitude sensors, including commanding, logging, and limit monitoring for wheel speed, temperature, and bus voltage.
- Support the mission data budget by estimating ADCS telemetry rates and volume for onboard storage and downlink planning.

Avionics Engineer, Ground Station & EPS Lead **Jan 2025 – Present**
THEIA (UMD Satellite Development Program CubeSat) *College Park, MD*

- Lead a 6-person Electrical Power System (EPS) team, owning subsystem architecture, power budgets, and Assembly, Integration, and Test (AIT) plans.
- Execute AIT procedures for spacecraft avionics, including the On-Board Computer (OBC) and S-band transmitter, in an ESD-safe cleanroom.
- Diagnosed an OBC-to-S-band pinout fault blocking RF transmission (unreachable TX-enable line); traced board schematics and built a custom remapping harness to restore link functionality.
- Engineered a 915 MHz UHF ground station transceiver using dual RFM9x LoRa modules and built the operator telemetry UI and automated OBC checkout suite.

Technical Projects

Custom Quadcopter Flight Computer & ESC · *STM32, Bare-Metal C, KiCad, Power Electronics*

- Design and route a 4-layer PCB flight computer integrating an STM32 MCU, 6-axis IMU over high-speed SPI, and regulated power rails with controlled impedance and split ground planes.
- Engineer a 3-phase brushless ESC using half-bridge N-channel MOSFET stages and gate drivers, optimizing high-current polygon pours to minimize switching-loop parasitic inductance.
- Write bare-metal C firmware using direct register access, DMA, and hardware timers for PID stabilization and PWM control.

Wearable Sensor Glove · *ROS2, C++, Python, Analog Sensing*

- Designed and prototyped a wearable glove with an array of analog flex sensors and IMUs to capture high-frequency ground-truth hand kinematics for ML pose validation.
- Implemented real-time ADC acquisition, sensor calibration, and voltage-to-joint-angle mapping algorithms.
- Developed low-latency ROS2 nodes in C++ and Python to stream synchronized multi-sensor telemetry.

Technical Skills

Languages: Matlab, C, C++, Python, SystemVerilog, Assembly
Protocols & RF: SPI, I2C, UART, RS-422/485, CAN, LoRa, CCSDS, RF Link Budgets
Hardware: 4-Layer PCB Design (KiCad), STM32, Power Electronics (MOSFETs, Buck/LDO), Op-Amps, Wire Harnessing, AIT
Test & Tools: Oscilloscopes, Logic Analyzers, Multimeters, Linux, ROS2, Git, Docker, SolidWorks, Jira, Confluence
Other: FCC Technician Class Amateur Radio License · English & Russian (native), Chinese, French, Spanish (conversational)